

# Immersive Bridge Inspection: Quantifying the Benefits of XR for Anomaly Detection

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## Abstract

We present an immersive bridge inspection tool as part of a digital twin for bridges that merges virtual reality (VR) with AI-assisted anomaly detection to evaluate how extended reality (XR) tools can modernize bridge inspections. Two tasks were studied: (i) relocating previously reported anomalies and (ii) detecting new surface anomalies using XR technologies. XR cues and simulated AI prompts were compared with manual inspection for five key performance indicators. Results show that XR + AI cut average detection time by 51%, boosted accuracy by 38%, lowered subjectivity by 11%, and reduced cognitive workload by 23 NASA-TLX points. A post-study survey yielded favorable usability and usefulness ratings, although some volunteers experienced VR-induced discomfort. These findings demonstrate the promise of XR and AI to accelerate, objectify, and lighten the mental load of structural assessments, while highlighting usability challenges that must be addressed for field deployment.

## CCS Concepts

• Human-centered computing → Virtual reality; Mixed / augmented reality.

## Keywords

Virtual reality, Augmented reality, Bridge inspection, Digital twin

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## 1 Introduction

Bridges are among the most critical pieces of infrastructure, and their failure can lead to severe financial losses and, more importantly, loss of life. Over time, bridges incur damage from natural

hazards, such as earthquakes and floods, and from long-term aging processes like corrosion and fatigue, which manifest as reduced load-bearing capacity and cracks. Traditional inspection methods are labor-intensive, time-consuming, and inherently subjective [1]. Structural Health Monitoring (SHM) addresses these challenges by deploying sensors, such as accelerometers and strain gauges, to continuously measure parameters such as vibration and strain, enabling early detection of damage and data-driven maintenance planning [4, 7].

Digital twin technology, ranked among the top ten emerging strategic technologies [5], offers a virtual replica of physical assets to support monitoring and prediction. Although digital twins have been adopted across various industries, their use in construction and infrastructure management remains limited [14]. Recent work demonstrates the potential of digital twins for predictive maintenance in civil structures [6]. Several studies have started exploring their application in construction workflows and bridge upkeep [2, 3, 9, 16]. To realize the full benefits of digital transformation, these virtual models should be integrated with cutting-edge technologies such as artificial intelligence and extended reality (XR) to enhance inspection accuracy and decision support [16]. To support this digital transformation, this paper explores the impact of XR technologies in bridge assessment and maintenance by designing and executing a virtual bridge inspection using Virtual Reality (VR).

One of the key technologies in a digital twin is XR visualization and interaction [16]. While the benefits of XR for bridge inspection and maintenance may appear intuitive, we explore the following research question (RQ): “What is the quantified impact of XR in bridge inspection?”. To answer this question, we designed a bridge inspection tool within a virtual environment and conducted experiments with users. The objectives of this experiment are as follows:

- Exploring the feasibility of VR for remote bridge inspection
- Developing a prototype application for remote inspection using VR as part of a digital twin for bridges
- Simulating and assessing the impact of AR and AI on visual on-site bridge inspection as part of a digital twin for bridges

We achieve these objectives by conducting virtual bridge inspection tasks under different conditions and measuring parameters, such as speed, accuracy, subjectivity, cognitive load, and user experience.

The remainder of this paper is organized as follows. Section 2 offers a short review of the state of the art in XR technologies



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for bridge inspection and maintenance. Section 3 presents our VR experiment designed to quantify the impact of XR on bridge inspection tasks. Section 4 reports and discusses the experimental results. In Section 5, we outline the limitations of our study and suggest directions for future work. Finally, Section 6 summarizes our key findings.

## 2 Related Work

XR techniques enable rich interaction with digital twins by visualizing sensor outputs, capturing user feedback, and facilitating collaboration among stakeholders. In particular, augmented reality (AR) supports the real-time visualization of sensor data and the real-time deployment of AI methods during on-site inspections, enhancing both efficiency and objectivity. For example, Samuel et al. [17] introduced the BIM and AR-based Supportive Inspection System (BASIS), which overlays historical defect information (type, size, and location) onto a bridge model. Their proof-of-concept on a pedestrian bridge reduced inspection time by 50%. VR, meanwhile, complements AR by providing fully immersive remote inspections, mitigating on-site hazards and enabling participation by experts who cannot be physically present.

While several studies have applied AR for remote inspection or 3D model visualization within digital twin and SHM frameworks [8, 11, 13, 15], VR remains underexplored in this context. We believe that VR's immersive capabilities are particularly well-suited to remote bridge inspection tasks. In this study, we empirically evaluate the impact of XR technologies, both AR and VR, on the accuracy, speed, cognitive load, and user experience of bridge inspection.

## 3 Method

### 3.1 Development of a VR Application for Remote Bridge Inspection

XR visualization and interaction are a key potential advantage of the digital twins for bridge inspection [16]. Thus, we designed and developed a VR-based bridge inspection tool to explore and quantify the impact of XR technologies in anomaly detection. The VR application was developed and tested on a Windows laptop equipped with 64GB of RAM and an NVIDIA RTX 4060 GPU. A Meta Quest 3 headset was also used for testing, along with a gaming-grade Wi-Fi router to ensure stable wireless streaming. The application was developed using Unity.

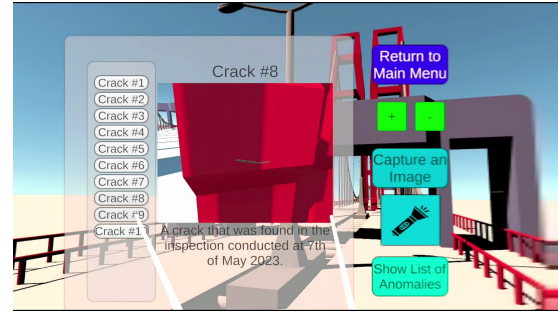
A 3D model of a realistic bridge [18] served as the test environment for the experiment. To simulate surface anomalies, crack images from a dataset [10] were processed and projected onto the bridge model, replicating typical visual defects encountered during inspections.

### 3.2 Experiment Design

Two tasks were simulated within the VR application:

- (1) Locating previously identified anomalies based on information from past inspection reports
- (2) Detecting new anomalies through a simulated visual inspection

**3.2.1 Locating the Previously Identified Anomalies.** Bridges typically have long operational lifespans, often extending over several



**Figure 1: The manual part of the VR experiment for locating the previously reported cracks.**

decades or even centuries. During this time, they undergo numerous inspections that document their structural condition. In particular, previously identified cracks and corrosion must be regularly monitored. However, relocating these documented anomalies during subsequent inspections can be both challenging and time-consuming. This task simulates a DT-powered AR/VR record-keeping system and evaluates its efficiency in identifying previously documented anomalies, in comparison to conventional manual visual inspection methods.

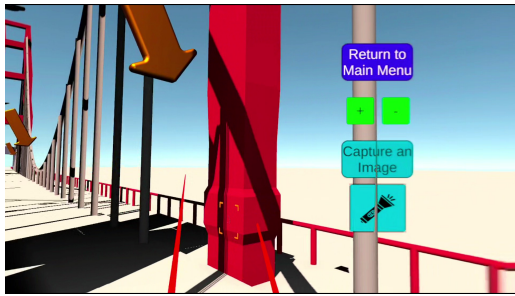
In this task, 10 cracks were projected onto the 3D model of the bridge, and a corresponding list, including images of each crack, was provided to the participant. The participant was then instructed to locate these cracks and capture images of them using the embedded camera within the application. The task was divided into two parts.

In the first part, the participant had to manually locate the cracks using only the information provided in the list, without any in-app visual assistance. This setup is illustrated in Figure 1. In the second part, cracks were highlighted using guiding arrows and bounding boxes, simulating an XR-assisted inspection environment. This setup is shown in Figure 2.

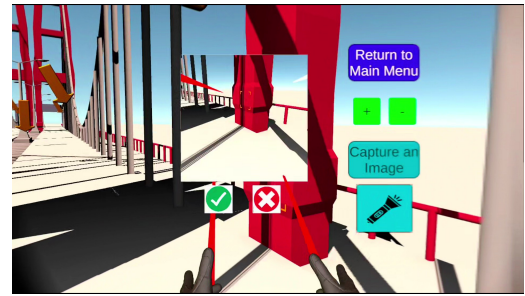
**3.2.2 Detecting New Anomalies.** Manual visual assessment remains the most common method for bridge inspection, though it is often challenging and subjective. This task evaluates the use of AI- and XR-powered surface anomaly detection in bridge inspections and compares the efficiency of identifying surface anomalies with and without the support of AI and XR technologies.

This task simulates the visual inspection of bridges, in which an inspector scans the structure to identify surface anomalies. A total of 10 cracks were projected onto the 3D model of the bridge, and participants were instructed to locate these cracks and capture their images using the in-application camera.

The task was divided into two parts. In the first part, participants conducted a manual visual inspection, relying solely on their observation skills to identify the cracks. In the second part, the inspection process was assisted by XR and AI technologies. When a participant approached a crack, a pop-up window appeared displaying an image of a candidate crack automatically detected by the AI module. The participant could then choose to accept or reject the suggested crack. The AI's role in this experiment was simulated through these manually crafted detection prompts. An example of such a pop-up window is shown in Figure 3b.

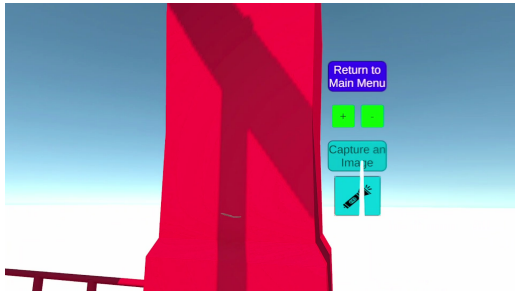


(a) A highlighted crack using XR

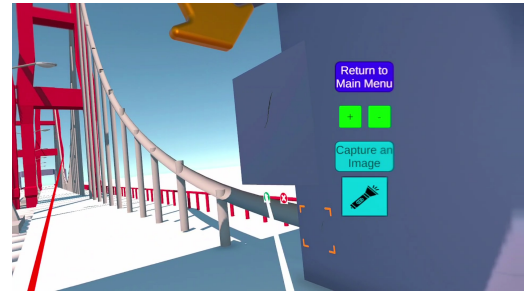


(b) Recording a located crack

Figure 2: The DT-enabled and XR-assisted visual inspection for locating previously reported cracks.



(a) Manual visual inspection in VR



(b) AI- and XR-assisted visual inspection

Figure 3: The DT-enabled visual inspection for detecting new cracks.

3.2.3 *Research Questions.* The following sub-research questions were formulated from the main research question (RQ):

- **RQ1:** What is the impact of the XR on the time required for visual inspection?
- **RQ2:** What is the impact of the XR on the accuracy of visual anomaly detection?
- **RQ3:** What is the impact of XR on the subjectivity of assessments?
- **RQ4:** What is the impact of XR on the cognitive load of the inspector?

### 3.3 Metrics and Evaluations

The following key performance indicators (KPIs) were used to evaluate the outcomes of the experiment:

- **KP1 - Time:** Average duration required to find an anomaly during an inspection task.
- **KP2 - Accuracy:** Average ratio of correctly detected anomalies during an inspection task.
- **KP3 - Subjectivity:** Standard deviation in performance among participants.
- **KP4 - Cognitive Load:** Average NASA Task Load Index (TLX) score reported for each task.
- **KP5 - User Experience:** Participant feedback collected via a Likert-scale survey evaluating the application's GUI.

While the KP1, KP2, and KP3 are straightforward, we believe KP4 and KP5 require further explanations, which are offered in the following subsections.

3.3.1 *Cognitive Load.* A standard method for quantifying task workload is the NASA Task Load Index (TLX) Questionnaire [12]. In this method, the workload index ranges from 0 to 100 and is calculated as the average of six dimensions: Mental demand, physical demand, temporal demand, performance, effort, and frustration level.

Each dimension is scored on a scale from 0 to 100, typically in 5-point increments. The dimensions are then paired into 15 unique combinations, and participants are asked to identify which dimension in each pair had a greater impact on their experience. These selections are used to assign weights to each dimension. The final workload index is computed as the weighted average of the six scores.

3.3.2 *User Experience.* In addition to the TLX assessment, participants were also asked to complete a Likert-scale survey at the end of the experiment, focusing on their user experience, as well as their perceptions of XR technologies in the context of bridge inspection.

A Likert-scale survey can be used to evaluate user experience and to gauge the perceived impact of XR technologies in bridge inspection from a subjective perspective. The survey consists of multiple statements, which participants rate on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree) at the end of the experiment.

## 4 Results and Discussion

A total of 14 volunteers participated in this experiment. Of these, 4 participants were unable to complete the experiment due to

headaches or nausea induced by operating in the virtual environment. The remaining 10 participants were evenly divided between two experimental tasks: Locating Previously Identified Cracks and Detecting New Cracks. All virtual sessions were recorded for further analysis, and a combination of surveys and questionnaires was used to assess cognitive load and user experience.

Prior to the main experiment, all participants completed a brief training session designed to familiarize them with the VR environment, user interface, navigation controls, inspection procedures, and the visual characteristics of cracks. Each participant performed their assigned task twice, once manually and once with AI/XR assistance. After completing each part of the task, participants filled out a NASA TLX Questionnaire to quantify their perceived workload.

To answer RQ1, Time (KP1) was measured. Table 1 presents the average detection time per anomaly and its standard deviation (STD) in seconds. The results show a reduction of approximately 56% and 51% in the time required for locating previously identified cracks and detecting new cracks, respectively—answering RQ1.

To answer RQ2, Accuracy (KP2) was evaluated. The average detection accuracy and its STD are also presented in Table 1. The results indicate an increase in accuracy of approximately 50% and 38% for locating previous anomalies and detecting new anomalies, respectively. The STD decreased by approximately 16% and 11%, suggesting more consistent performance across participants. These findings answer RQ2. Furthermore, the reduction in standard deviation also answers RQ3, which shows a reduction in subjectivity (KP3) through XR assistance.

To explore RQ4, workloads measured via the NASA TLX are reported in Table 2. The results show that the weighted average workload decreased by approximately 29 points for the task of locating previous anomalies and 23 points for detecting new anomalies when comparing XR-enabled inspection with manual methods. These findings answer RQ4.

Lastly, the Likert-scale survey consisted of 13 statements related to user experience and the perceived impact of XR technologies in bridge inspection. These statements were rated by participants on a five-point scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree) and were completed at the end of the experiment.

**Table 1: Impact of the AI-/XR-assisted visual inspection compared to manual visual inspection.**

| Metric                     | Task 3.2.1 |      | Task 3.2.2 |       |
|----------------------------|------------|------|------------|-------|
|                            | Manual     | XR   | Manual     | AI&XR |
| Average detection time (s) | 53         | 23   | 43         | 21    |
| STD of detection time (s)  | 12         | 7    | 17         | 7     |
| Average accuracy (%)       | 46         | 96   | 52         | 90    |
| STD of accuracy (%)        | 21.91      | 5.48 | 25.88      | 14.14 |

**Table 2: Impact of the AI-/XR-assisted inspection on cognitive load.**

| Category                  | Task 3.2.1 |      | Task 3.2.2 |       |
|---------------------------|------------|------|------------|-------|
|                           | Manual     | XR   | Manual     | AI&XR |
| Mental Demand             | 35         | 9    | 28         | 12    |
| Physical Demand           | 35         | 20   | 13         | 9     |
| Temporal Demand           | 47         | 28   | 42         | 18    |
| Performance               | 50         | 8    | 33         | 8     |
| Effort                    | 43         | 8    | 37         | 13    |
| Frustration Level         | 35         | 18   | 29         | 11    |
| Average workload          | 40.8       | 15.1 | 30.3       | 11.8  |
| Weighted average workload | 44.0       | 14.0 | 36.4       | 13.1  |

The results indicate that participants generally agreed that the user experience was positive, with an average score of 4. Furthermore, the average score for statements regarding the positive impact of XR in bridge visual inspection was also above 4, suggesting that most participants see XR-assisted inspection as a promising direction for the future.

Interestingly, the survey also revealed that participants did not express a strong preference between using dedicated XR headsets/glasses and handheld devices (e.g., smartphones or tablets) for XR-assisted inspection, with preferences appearing to be evenly distributed.

From a technical standpoint, XR headsets/glasses offer several advantages over handheld devices, including a wider field of view and hands-free operation, both of which are critical for ensuring the safety and efficiency of inspectors during on-site inspections. These findings suggest that, while technically superior, XR headsets may require greater effort in terms of usability, training, and integration strategies to gain broader acceptance in industry settings.

## 5 Limitations and Future Directions

Approximately 28% of participants experienced nausea or headaches with VR, revealing a significant usability barrier for bridge inspection applications. Participants familiar with AR reported lower cognitive load than with VR, indicating that AR-focused tools may offer broader accessibility. To reduce motion sickness, future VR systems should provide adjustable movement and rotation speeds, alternative locomotion methods (e.g., teleportation or blink), and floating movement modes to minimize bobbing. Interface improvements are also needed: HUD elements must remain readable in peripheral view and avoid occlusion by virtual objects.

Finally, the study's small sample size and fixed task order (manual inspection followed by XR) may introduce learning effects; future work should involve larger, counterbalanced cohorts and include comparisons to physical inspections.

## 6 Conclusion

A virtual inspection experiment in VR showed that incorporating XR technologies can increase inspection speed and accuracy by 51% and 38%, respectively. At the same time, the XR-assisted inspection

process led to a reduction in subjectivity by 11% and a decrease in cognitive workload by 23 points, as measured by the NASA TLX index.

These findings demonstrate that XR-enhanced virtual inspections can substantially accelerate structural assessments and improve defect detection while easing both bias and mental effort. User feedback further indicates strong acceptance of immersive XR-based digital twin interfaces for bridge inspection tasks.

Nonetheless, issues such as motion sickness and potential learning effects warrant further exploration of alternative locomotion methods, adaptive comfort settings, and intuitive designs. Additionally, in situ AR trials are necessary for validating the impact of AR. Overall, this work highlights the promise of XR + AI platforms to transform infrastructure maintenance through faster, more reliable, and less strenuous inspection workflows.

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## References

- [1] A. Emin Aktan, Daniel N. Farhey, David L. Brown, Vikram Dalal, Arthur J. Helmicki, Victor J. Hunt, and Stuart J. Shelley. 1996. Condition Assessment for Bridge Management. *Journal of Infrastructure Systems* 2, 3 (Sept. 1996), 108–117. doi:10.1061/(ASCE)1076-0342(1996)2:3(108) publisher: American Society of Civil Engineers.
- [2] Kazi Masudul Alam and Abdulmotaheb El Saddik. 2017. C2PS: A Digital Twin Architecture Reference Model for the Cloud-Based Cyber-Physical Systems. *IEEE Access* 5 (2017), 2050–2062. doi:10.1109/ACCESS.2017.2657006
- [3] Didem GÜRDÜR Broo, Miguel Bravo-Haro, and Jennifer Schooling. 2022. Design and implementation of a smart infrastructure digital twin. *Automation in Construction* 136 (4 2022), 104171. doi:10.1016/j.autcon.2022.104171
- [4] Fikret Necati Catbas, Furkan Luleci, Mahta Zakaria, Ulas Bagci, Joseph J. LaViola, Carolina Cruz-Neira, and Dirk Reiners. 2022. Extended Reality (XR) for Condition Assessment of Civil Engineering Structures: A Literature Review. *Sensors* 22 (12 2022), 9560. Issue 23. doi:10.3390/s22239560
- [5] David Cearley, Brian Burke, Alfonso Velosa, and Marc Kerremans. 2019. Top 10 Strategic Technology Trends for 2019: Digital Twins.
- [6] Itxaro Errandonea, Sergio Beltrán, and Saioa Arrizabalaga. 2020. Digital Twin for maintenance: A literature review. *Computers in Industry* 123 (12 2020), 103316. doi:10.1016/J.COMPIND.2020.103316
- [7] Charles R Farrar and Keith Worden. 2006. An introduction to structural health monitoring. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 365, 1851 (Dec. 2006), 303–315. doi:10.1098/rsta.2006.1928 publisher: Royal Society.
- [8] Muhammad Fawad, Marek Salamat, Muhammad Usman Hanif, Kálmán Koris, Muhammad Ahsan, Hadiya Rahman, Michael Gerges, and Mostafa M Salah. 2024. Integration of bridge health monitoring system with augmented reality application developed using 3D game engine—Case Study. *IEEE Access* (2024).
- [9] Enxhi Ferko, Alessio Bucaioni, and Moris Behnam. 2022. Architecting Digital Twins. *IEEE Access* 10 (2022), 50335–50350. doi:10.1109/ACCESS.2022.3172964
- [10] Tao Jin and Zhexun Li. 2022. Bridge Crack Library 2.0. doi:10.7910/DVN/TUFAJT
- [11] Katrina Mae Montes, Ji Dang, Jiaming Liu, and Pang-jo Chun. 2023. Bridge Status Realization and Management Enhanced by UAV, SfM, and Deep Learning. In *European Workshop on Structural Health Monitoring*, Piervincenzo Rizzo and Alberto Milazzo (Eds.), Springer International Publishing, Cham, 536–545.
- [12] NASA. 1986. NASA task load index (TLX).
- [13] Duy-Cuong Nguyen, The-Quan Nguyen, Ruoyu Jin, Chi-Ho Jeon, and Chang-Su Shim. 2022. BIM-based mixed-reality application for bridge inspection and maintenance. *Construction Innovation* 22 (6 2022), 487–503. Issue 3. doi:10.1108/CI-04-2021-0069
- [14] De Graft Joe Opoku, Srinath Perera, Robert Osei-Kyei, and Maria Rashidi. 2021. Digital twin application in the construction industry: A literature review. *Journal of Building Engineering* 40 (8 2021), 102726. doi:10.1016/J.JOBE.2021.102726
- [15] Neha Patel, Komal Parikh, and Bhagyesh Patel. 2021. Bridge information modeling and AR using terrestrial laser scanner. *Reliability: Theory and Applications* 16 (2021), 17 – 23.
- [16] Ali Kh. Sabzi, Berend Jan van der Zwaag, Duc V. Le, and Özlem Durmaz-Incel. 2025. Digital Twins for Bridge Assessment and Maintenance. *International Conference on Smart Applications, Communications and Networking (SmartNets)* (2025).
- [17] Immanuel John Samuel, Ossama Salem, and Song He. 2022. Defect-oriented supportive bridge inspection system featuring building information modeling and augmented reality. *Innovative Infrastructure Solutions* 7 (8 2022), 1–17. Issue 4. doi:10.1007/S41062-022-00847-3
- [18] SK-Blendz. 2021. Realistic Bridge over body of water free 3D model. <https://www.cgtrader.com/items/3263428/download-page>